UNIVERSALLY BOUNDED FIBRES COMPOSE: THE BOUND MULTIPLIES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 305-324, immediately following the statement at 296-303.

Editorial note. One cross-chapter pointer (Morphisms) is recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Definition of universally bounded fibres. Definition definition-universally-bounded of this chapter, cited at proof lines 305-307. Fixes the bounds n (for V -> U) and m (for U -> X) used in the composition.
- Degree bounds compose on schemes. Morphisms, Lemma morphisms-lemma-composition-universally-bounded, cited at proof lines 317-319. The imported scheme-side composition: finite of degree <= m composed with fibres of degree <= n gives degree <= nm.

Source reference. Original definition 03JL: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 277–277. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03J6: focused-seeds/geometry/S9-decent-composition-universally-bounded/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $V \to U$ be a morphism of schemes over S, and let $U \to X$ be a morphism from U to X. If the fibres of $V \to U$ and $U \to X$ are universally bounded, then so are the fibres of $V \to X$.*

Proof. Let n be an integer which works for $V \to U$, and let m be an integer which works for $U \to X$ in Definition 03JL. Let $\mathrm{Spec}(k) \to X$ be a morphism, where k is a field. Consider the morphisms

$$\mathrm{Spec}(k) \times_X V \longrightarrow \mathrm{Spec}(k) \times_X U \longrightarrow \mathrm{Spec}(k).$$

By assumption the scheme $\mathrm{Spec}(k) \times_X U$ is finite of degree at most m over k, and n is an integer which bounds the degree of the fibres of the first morphism. Hence by Morphisms, Lemma 03J6 we conclude that $\mathrm{Spec}(k) \times_X V$ is finite over k of degree at most nm. □

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